Final Product Change Notification 201609007F01

Issue Date: 02-Oct-2016
Effective Date: 01-Apr-2017

Dear *Product Data*,

Here's your personalized quality information concerning products Mouser Electronics purchased from NXP.
For detailed information we invite you to view this notification online

Management Summary

- Replace Au (gold) wire with Cu (copper) wire in bonding process
- Add second source for NXP ICN8 fab (Nijmegen, The Netherlands) C075 and C125 processes

Change Category

☐ Wafer Fab Process	☑ Assembly Process	☑ Product Marking	☐ Test Location	☐ Design
☐ Wafer Fab Materials	☑ Assembly Materials	☐ Mechanical Specification	☐ Test Process	☐ Errata
☑ Wafer Fab Location	☐ Assembly Location	☐ Packing/Shipping/Labeling	☐ Test Equipment	☐ Electrical spec./Test coverage

Introduction of Cu-wire bonding and ASMC second source for automotive products

Details of this Change

Introduction of Cu-wire bonding and ASMC as front end second source for selected automotive logic products

- Replace Au (gold) wire with Cu (copper) wire in bonding process
- Add second source for NXP ICN8 fab (Nijmegen, The Netherlands) C075 and C125 processes

Affected products and locations:
- TSSOP5/6 products from ATSN (NXP Semiconductors Assembly & Test Plant Seremban Malaysia)
- SO and TSSOP products from ATBK (NXP Semiconductors Assembly & Test Plant Bangkok, Thailand) and ASEN (NXP-ASE JV Suzhou, China)
- Second source LVC, AUP and AVC products processed in ASMC (Advanced Semiconductor Manufacturing Corporation Limited in Shanghai China)

Qualification in accordance to the Automotive Electronics Council:
- AEC-Q100-rev. H Stress Test Qualification for Integrated Circuits
- AEC-Q006-rev. A Qualification requirements for Cu wire interconnections

Impact:
- Matching wafer PCM (Process Control Modules) and product parameters by 1.5 delta/sigma
- No change in die layout, all masks are produced from the same GDS2 file
- No change in form, fit, function, quality or reliability

Why do we Implement this Change

- Next phase in Cu wire expansion after maturation in commodity logic products since 2012
- Continued alignment with world technology trends on state of the art production tools
- Copper wire shows enhanced mechanical properties

- Increased environmental friendliness (eco-friendly)
- Front End capacity expansion and add second source for critical logic processes

Identification of Affected Products

- Assembly changes can be identified by backward traceability of the product marking date code
- Applied wafer fab is indicated on the reel, box label and on the product marking (if space permits)

Product Availability

Sample Information

Samples are available upon request
Samples are available upon request from the Logic sample store Nijmegen The Netherlands

Production

Planned first shipment 01-Apr-2017

Impact

- No impact to the product's functionality or parametric spread anticipated

Data Sheet Revision

No impact to existing datasheet

Disposition of Old Products

Existing inventory will be shipped until depleted

Related Notifications

Notification	Issue Date	Effective Date	Title
201202018F01	05-Dec-2012	05-Mar-2013	Cu-wire bonding in APB and ASEN for SO14-16 packages
201208012F01	05-Dec-2012	05-Mar-2013	Cu-wire bonding in APB and ASEN for TSSOP14-16-20 packages
201304005F01	17-Apr-2013	16-Jul-2013	Cu-wire bonding in APB and ASEN for SO14-16 packages (Phase II)
201304006F01	17-Apr-2013	16-Jul-2013	Cu-wire bonding in APB and ASEN for TSSOP14-16-20 packages (Phase II)
201401003F01	15-Jun-2014	13-Sep-2014	Cu-wire bonding in APB and ASEN for TSSOP48-56 packages (Phase III)
201409013F01	09-Jan-2015	24-Apr-2015	Introduction of Cu-wire bonding for TSSOP5 and DHVQFN20 packages (Phase IV)
201410001F01	01-Nov-2014	13-Feb-2015	Introduce ASMC as a second source wafer fab location for 74AUP Logic products
201412011F01	11-Jan-2015	24-Apr-2015	Introduction of Cu-wire bonding for SO20 package (Phase IV)
201503008F01	26-Mar-2015	22-Jul-2015	Phase 2: Expansion of industrial footprint for C075 process 74AVC/AUP Logic products
201505005F01	12-Sep-2015	25-Dec-2015	Phase 3: Expansion of industrial footprint for C075/C125 process 74LVC/AVC/AUP (ASMC)
201605004F01	30-Jun-2016	14-Oct-2016	Expansion of Cu-wire bonding for TSSOP5/6 packages (Phase V)
201605005F01	30-Jun-2016	14-Oct-2016	Introduction of Cu-wire bonding for SO20-24 and TSSOP24 packages in ATBK (Phase VI)

Additional information

Self qualification: view online

Timing and Logistics

Your acknowledgement of this change, conform JEDEC JESD46 D, is expected till 31-Oct-2016.

Remarks

Wafer Fab Indicators:
"Y"= ASMC, Shanghai, China
"T"= NXP ICN8, Nijmegen, Netherlands

Contact and Support

For all inquiries regarding the ePCN tool application or access issues, please contact NXP "Global Quality Support Team".

For all Quality Notification content inquiries, please contact your local NXP Sales Support team.

For specific questions on this notice or the products affected please contact our specialist directly:

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Position	PCN specialist - QA engineering Logic Products
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Customer Focus, Passion to Win.

NXP Quality Management Team.

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